



EEHC DISTRIBUTION MATERIALS SPECIFICATION

EDMS 04- 300 - 1

Date: 21-10-2018

EDMS 04-300-1

SPECIFICATION

FOR

SUPPLY OF LOW VOLTAGE INSULATED

CONDUCTORS

Issue: Oct-2018 / Rev- 1



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I- SCOPE:

This specification covers the design , manufacture , testing and supply of 0.6/1KV overhead aluminum or copper stranded wires conductors with XLPE insulation with carbon black content

II- GENERAL DESCRIPTION_:

These insulated conductors should be used for installation in low voltage networks .

The design and construction of insulated conductors shall secure the followings:

- a) Maximum electrical strength.
- b) Maximum mechanical strength.
- c) Minimum conductor resistance.
- d) Maximum flexibility i.e. minimum bending radius.
- e) Maximum thermal conductivity i.e. maximum heat dissipation.
- f) High resistance against water penetration , sunrays , sand storm , acids, alkalines and salts .

The maximum insulated conductor temperature shall be rated for 90oC continuous temperature and 130oC maximum emergency temperature and 250oC short-circuit temperature.



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III- APPLICABLE STANDARDS :-

In addition to the requirements of this specification, all conductors should comply with the latest relevant applicable standards issued by the INTERNATIONAL ELECTROTECHNICAL COMMISSION (IEC) IEC60502-1, IEC60889, IEC60811, IEC60538, IEC61089, IEC60228 , or the latest edition of any other equivalent specifications .

IV-ENVIRONMENTAL CONDITIONS:

- The insulated conductors shall be designed to meet the following requirements.
 - a) Ambient temperature at point of use : -5oC to 55o C.
 - b) Relative humidity : 10% to 100%.
 - c) Occasional sand storms.
 - d) Salty air.
 - e) sun rays exposure.

III- CONSTRUCTION:-

A- CONDUCTORS:-

The conductor should be class 2, plain aluminum (H14) or cooper, stranded of circular shape. The minimum number of wire strands in the conductor shall be as follows



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No .	Conductor nominal cross-sectional area (mm ²)	Minimum number of wire strands in conductor	
		Aluminum	Copper
1	16	7	7
2	25	7	7
3	35	7	7
4*	50	19 (7 as option)	19 (7 as option)
5	70	19	19
7	95	19	19
8	120	37	37
9	150	37	37

* Option if required byEDC

The maximum resistance of conductor at 20°C shall not exceed the specified value in the latest relevant applicable IEC standards.

B- INSULATION:-

- The insulation shall be cross-linked polyethylene - (2.5 ± 0.5) % carbon black content and resist of ultra-violet rays.
- The thickness of insulation layer should be not less than the (specified value) as follows: -



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Conductor Cross Sectional area (mm ²)	Nominal thickness of Insulation specified value (mm)
16	0.7
25	0.9
35	0.9
50	1
70	1.1
95	1.1
120	1.2
150	1.4

The insulation resistance at 90o C shall be not less than the specified value in the latest relevant applicable IEC standards.

IV- INSPECTION AND TESTING :-

- 1- Distribution Company (..... EDC) reserves its right to carry out Inspection during fabrication stages and witnesses the testing of the conductors.
- 2- Unless otherwise specified or approved in writing by EDC, the conductors shall be subjected to inspection and type tests in accordance with all relevant applicable IEC standards.
- 3- Testing methods, selection of tests' samples and rejection of conductors shall be in accordance with all applicable IEC standards.



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- 4- All tests shall be carried out by the manufacturer and at his expense and under control and supervision of EDC inspection engineers.
- 5- Type tests' certificates shall be delivered to EDC.
- 6- Attendance of EDC representatives during fabrication stages and testing shall not relieve the manufacturer of his full responsibility for furnishing the conductors in accordance with the requirements of this specification and shall not give him the right to invalidate any claim that EDC may make because of defective or unsatisfactory material or faulty workmanship.

V- MARKING:-

- The marking of insulated conductors shall be in accordance with IEC standards.
- On each reel the following data shall be printed on both sides and could not be erased easily:

For: EDC

Manufactured by:.....

Year of manufacture:.....

Voltage class: KV

Material of conductor:.....

Material of insulation:.....

cross-sectional area:.....

Length of conductor on reel: m

Net weight: Kg

Gross weight: Kg

Reel No/ date of manufacture.....



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The insulated conductor shall be marked throughout each meter of its length with the following items:

1. Voltage rating KV
2. cross-sectional area mm²
3. Manufacturer's name
4. Year of manufacture
5. Customer's name (..... EDC)
6. Insulation type
7. Conductor material

VII-MISCELLANEOUS:-

- The insulated conductor on each wooden reel shall be as long as possible, and not be less than 1000 meters.
- The conductor ends shall be firmly and properly secured to the reel and shall be covered with sealing end caps.
- The insulated end seals shall be installed in the factory to insure that the cable remains water tight from the manufacturing up to the installation.
- The tenderer shall show in his tender the length of cable be supplied as one length on each reel which shall be maximum that can be placed on the reel without damaging the cable or causing deformation of the reel (drum).
- The charged reels shall be covered with wooden slabs of suitable thickness.
- If EDC required , Steel tape should be wrapped around the wood slabs for each side of the reel



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VIII- submittals:-

The tenderer should submit with his offer the following (at least)

- 1- three copies of detail drawings of conductors.
- 2- Original catalogue
- 3- instructions illustrating and explaining design criteria and bases
- 4- reference standards applied to the conductors
- 5- instructions for erection, operation and maintenance.
- 6- quality control procedure.

X- TESTS

1) Type tests:

- a) joints in aluminum wires.
- b) stress-strain curves.
- c) breaking strength of conductor.

2) Sample size:

Samples for the tests shall be taken at random from the outer end of not less than 10% of the drums of conductor. However, the inspection of the surface condition of the conductor shall be carried out on every drum prior to lagging.

3) Sample length:

3.1) Samples for tests of individual wires after stranding when requested, shall consist of a 1.5 m length or cut from the outer end of the coils or drums of conductors.

3.2) The sample length required for tensile and stress-strain tests shall be at least 400 times the diameter of the conductor but not less than 10 m.



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4) Sample tests:

4.1) Cross-sectional area.

The cross-sectional area shall not vary from the nominal value by more than $\pm 2\%$ in any sample and by more than $\pm 1.5\%$ for the average of any four measured values at locations selected at random with a minimum spacing of 20 cm.

4.2) Conductor diameter:

The diameter of the conductor shall not vary by:

$\pm 1\%$ for diameters larger than or equal to 10 mm;

± 0.1 mm for diameters smaller than 10 mm.

4.3) Carbon black content ($2.5 \pm 0.5\%$) test.

4.4) D.C. resistance of conductor at 20 °C.

4.5) Hot set test.

5) Acceptance or rejection

5.1 Failure of a test specimen to comply with any one of the requirements of this standard shall constitute grounds for rejection of the lot represented by the specimen.

5.2 If any lot is so rejected, the manufacturer shall have the right to test, only once, all individual drums of conductors in the lot and submit those which meet the requirements for acceptance.

IX- GURANTEE DATA:-

The tenderer shall fill in thoroughly in type-writing the attached guarantee data for the offered conductors.



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GUARANTEE DATA OF INSULATED CONDUCTORS

A - PARTICULARS:-

- Maker's name :.....
- Standard applied :.....
- Nominal voltage :.....
- Shape of conductor :.....
- Number of conductor strands :.....
- Diameter of strands :.....mm
- Cross-sectional area of strands :.....mm²
- Overall diameter of conductor :.....mm
- Cross-sectional area of conductor:.....mm²
- Material of conductor:.....
- Material of conductor insulation :.....
- Thickness of conductor insulation :
 - * Nominal value :.....mm
 - * Minimum value :.....mm
 - * Maximum value :.....mm
- Overall insulated conductor diameter :.....mm
- Conductor weight per km :
 - * Aluminium / copper :..... kg
 - * Finished conductor :..... kg



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- Length of conductor per reel :.....m
- Reel diameter :.....m
- Reel width :.....m
- Gross weight of charged reel :.....kg

B - PERFORMANCE :-

- Minimum bending radius :.....cm
- Thermal conductivity :..... w/cm.c
- Maximum d.c. resistance of conductor at 20 °C per km :..... ohm
- Maximum a.c. resistance of insulation at 90 °C per km :..... ohm
- Inductance per km :..... ohm
- Reactance at 50 c/s per km :..... ohm
- Impedance at 70 °C and 50 c/s per km :..... ohm
- Maximum continuous current rating :
 - * IN air 45 C :.....Amps
 - * Inconduit :.....Amps
- Maximum one second symmetrical short circuit current :..... °C
- Maximum cable temperature :
 - * Continuous:..... °C
 - * Emergancy :..... °C
 - * Short-circuit :..... °C

I / we certify that the above-mentioned data are guaranteed for the offered insulated conductor.

Signature :.....

Date :.....